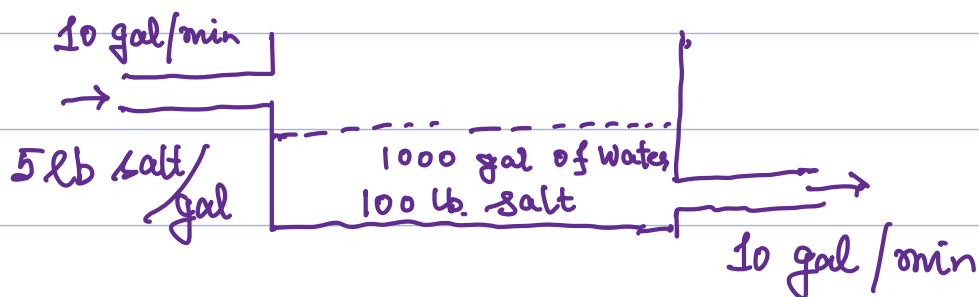


MTH 204 : Lecture 02

Mixing Problem: A tank contains 1000 gal of water in which initially 100 lb of salt is dissolved. Brine runs in at a rate of 10 gal/min and each gallon contains 5 lb of dissolved salt. The mixture in the tank is kept uniform by stirring. Brine runs out at 10 gal/min. Find the amount of salt in the tank at any time t .



To find the amount of salt at any time.

Let $Y(t)$ be the amount of salt at time t .

Given: $Y(0) = 100$

$$\frac{dy}{dt} = \underbrace{\text{salt inflow rate}} - \underbrace{\text{salt outflow rate}}$$

$$\Rightarrow \frac{dy}{dt} = 5 \times 10 \text{ lbs/min} - \frac{Y}{1000} \times 10 \text{ lbs/min}$$

$$\Rightarrow \frac{dy}{dt} = \frac{5000 - y}{100}$$

$$\Rightarrow \frac{dy}{y-5000} = -\frac{dt}{100}$$

$$\Rightarrow \ln |y-5000| = -0.01t + C_1$$

$$\Rightarrow |y-5000| = Ce^{-0.01t} \quad \text{where } C = e^{C_1}$$

$$\text{Now } y(0) = 100 \Rightarrow |100-5000| = Ce^0 \Rightarrow C = |-4900| = 4900$$

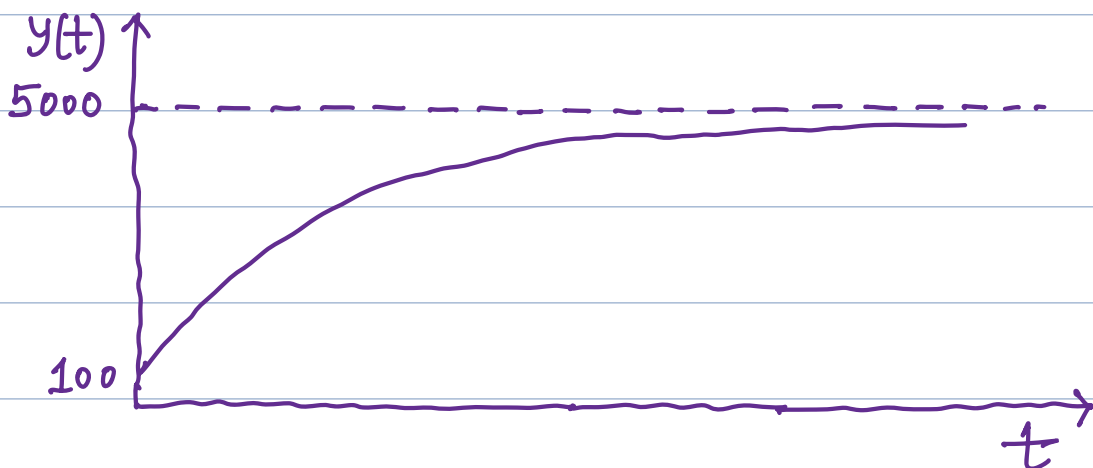
$$\text{Hence } |y-5000| = 4900e^{-0.01t}$$

Now from the problem, we can see that $y(t) < 5000$

$$\text{So, } |y-5000| = -(y-5000) = 4900e^{-0.01t}$$

$$\Rightarrow 5000 - y = 4900e^{-0.01t}$$

$$\text{So, } \boxed{y(t) = 5000 - 4900e^{-0.01t}}$$



Ex: (Newton's Law of Cooling):

Suppose that in the winter, the daytime temperature is maintained at 70°F (in a building). The Heating is turned off at 10 PM and turned on again at 6 AM.

On a certain day, the temperature inside the building at 2 AM was found to be 65°F .

The outside temperature was 50°F at 10 PM and had dropped to 40°F by 6 AM.

What was the temperature inside the building at 6 AM?

Physical information:

Newton's Law of Cooling states that the time rate of change of temperature T of a body is proportional to the difference between T and the temperature of the surrounding medium.

	<u>10 PM</u>	<u>2 AM</u>	<u>6 AM</u>
$\left(70^{\circ}\right)$ (inside)	50° (outside)	65° (inside)	40° (outside)

Let $T(t)$ be the temperature inside the

building at time t (after 10 PM).

To get the temperature outside at any time t , we take an average

$$T_{\text{out}} = \frac{50+40}{2} = 45^\circ \text{F}$$

By Newton's Law of Cooling

$$\frac{dT}{dt} = k(T - T_{\text{out}})$$

$$\Rightarrow \frac{dT}{dt} = k(T - 45)$$

$$\Rightarrow \frac{dT}{T-45} = k dt$$

$$\text{Then } T-45 = C e^{kt}$$

$$\Rightarrow T(t) = 45 + C e^{kt}$$

$$\begin{aligned} \text{given: } T(0) &= 70 \Rightarrow 70 = 45 + C \\ (\text{at 10 PM}) &\Rightarrow C = 25 \end{aligned}$$

$$\Rightarrow T(t) = 45 + 25 e^{kt}$$

$$\text{When } T(4) = 65^\circ \text{ (at 2 AM)}$$

$$\Rightarrow 65 = 45 + 25 e^{k(4)}$$

$$\Rightarrow 20 = 25 e^{k(4)}$$

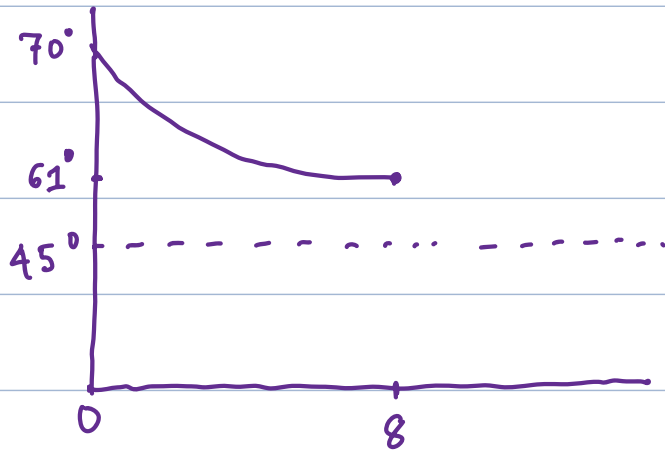
$$\Rightarrow \frac{4}{5} = e^{k(4)} \Rightarrow k = \frac{1}{4} \ln\left(\frac{4}{5}\right)$$

$$\approx -.056$$

So, $T(t) = 45 + 25e^{-.056t}$

Want T at 6 AM

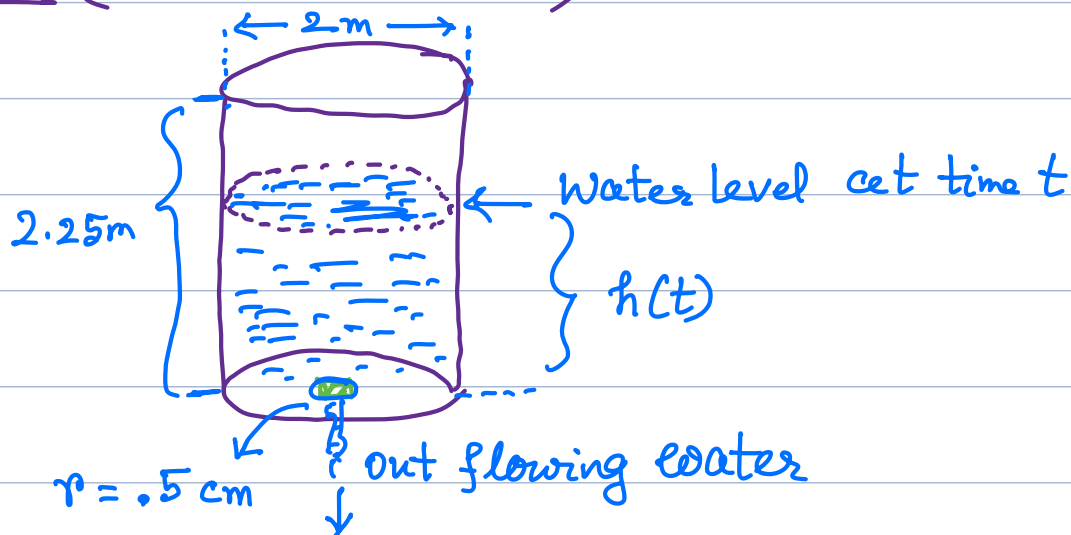
i.e. at $t = 8$



$$T(8) = 45 + 25e^{-.056 \times 8}$$

$$= 60.97 \approx 61^\circ \text{F}$$

Ex: (Torricelli's Law):



Physical information: Under the influence of gravity

the out flowing water has velocity

$$v(t) = .6 \sqrt{2gh(t)}$$

where $h(t)$ is the height of the water above the hole at time t
and $g = 980 \text{ cm/sec}^2 = 32.17 \text{ ft/sec}^2$ is the acceleration due to gravity at the surface of the earth.

Let ΔV be the volume of water outflowing through a hole in time Δt and let A be the surface area of the hole.

$$\text{Then } \Delta V = A v \Delta t = A v(t) \Delta t$$

$$\text{Also, } \Delta V \propto \Delta h \Rightarrow \Delta V = -B \Delta h \text{ where } B \text{ is the base surface area}$$

$$\Rightarrow -B \Delta h = A v \Delta t$$

$$\Rightarrow \frac{\Delta h}{\Delta t} = -\frac{A}{B} v$$

Taking limit as $\Delta t \rightarrow 0$ we get

$$\frac{dh}{dt} = -\frac{A}{B} v = -\frac{A}{B} (.6) \sqrt{2gh(t)}$$

$$\Rightarrow \frac{dh}{dt} = D \sqrt{h} \text{ where } D = -\frac{A}{B} (.6) \sqrt{2g}$$

$$\Rightarrow \frac{dh}{\sqrt{h}} = D dt$$

$$\int h^{-\frac{1}{2}} dh = \int D dt$$

$$\Rightarrow 2h^{\frac{1}{2}} = Dt + C_1$$

$$\Rightarrow \sqrt{h} = \frac{D}{2}t + C \quad \left(C = \frac{C_1}{2}\right)$$

$$\Rightarrow \boxed{h = \left(\frac{D}{2}t + C\right)^2}$$

Now, $D = -\frac{A}{B} \cdot 6\sqrt{2g}$

$A = \pi(100)^2 \text{ cm}^2$
 $B = \pi(.5)^2 \text{ cm}^2$

Then $D = -.000664$

$$h = \left(-.000332t + C\right)^2 \text{ cm.}$$

$$h(0) = 2.25 \text{ m} = 225 \text{ cm}$$

$$225 = C^2 \Rightarrow C = 15$$

So, $\boxed{h = \left(-.000332t + 15\right)^2 \text{ cm}}$

Reduction to Separable form:

For those ODEs that can be written as:

$$y' = f(y/x)$$

we make the change of variable

$$u = \frac{y}{x}$$

$$\Rightarrow y = ux \Rightarrow y' = u'x + u$$

$$\text{Then } y' = f(y/x) \Rightarrow u'x + u = f(u)$$

$$\Rightarrow u'x = f(u) - u$$

$$\Rightarrow x \frac{du}{dx} = f(u) - u$$

$$\Rightarrow \frac{du}{f(u) - u} = \frac{dx}{x}$$

Now the equation is reduced to a separable form and can be integrated now.

Ex: $2xyy' = y^2 - x^2$

$$\Rightarrow y' = \frac{y^2}{2xy} - \frac{x^2}{2xy}$$

$$\Rightarrow y' = \frac{1}{2} \frac{y}{x} - \frac{1}{2} \frac{x}{y}$$

$$\Rightarrow y' = \frac{1}{2} \left(\frac{y}{x} - \frac{x}{y} \right)$$

$$\text{Let } u = \frac{y}{x} \Rightarrow y = ux \Rightarrow y' = xu' + u.$$

$$\text{So, } xu' + u = \frac{1}{2} \left(u - \frac{1}{u} \right)$$

$$\Rightarrow xu' = -u + \frac{1}{2}u - \frac{1}{2u}$$

$$\Rightarrow xu' = -\frac{1}{2}u - \frac{1}{2u} = -\frac{1}{2} \frac{(u^2 + 1)}{u}$$

$$\Rightarrow \int \frac{2u \, du}{u^2 + 1} = - \int \frac{dx}{x}$$

$$\Rightarrow \ln(u^2 + 1) = -\ln|x| + C_1$$

$$\Rightarrow u^2 + 1 = \frac{C}{x}$$

$$\Rightarrow \frac{y^2}{x^2} + 1 = \frac{C}{x}$$

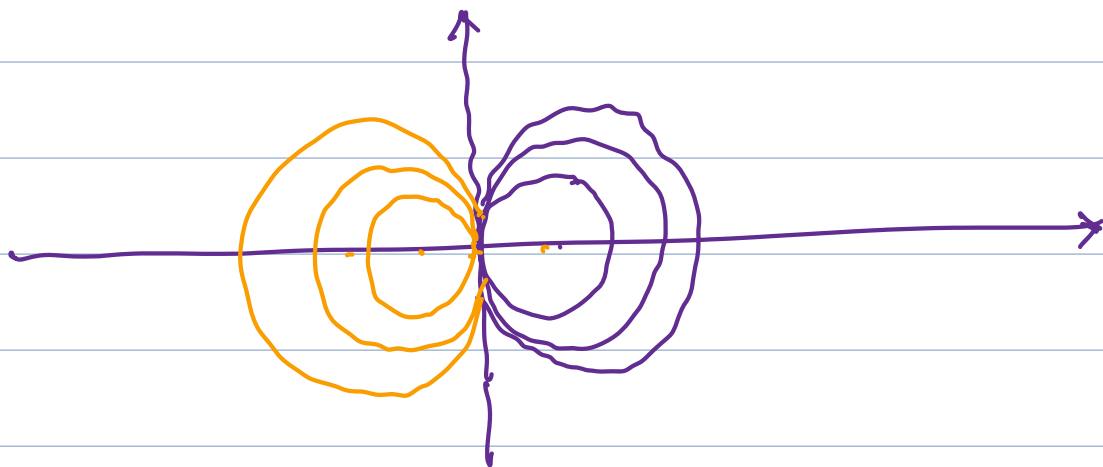
$$\Rightarrow \frac{y^2 + x^2}{x^2} = \frac{C}{x} \Rightarrow x^2 + y^2 = Cx$$

$$\Rightarrow \left(x - \frac{C}{2} \right)^2 + y^2 = \frac{C^2}{4}$$

Thus the general solution is the

family of circles centered at $\left(\frac{C}{2}, 0 \right)$ and

radius $\frac{|C|}{2}$



General solution (family of circles)

Ex: $xy' = y + 2x^3 \sin^2\left(\frac{y}{x}\right)$

$$\Rightarrow y' = \frac{y}{x} + 2x^2 \sin^2\left(\frac{y}{x}\right)$$

Let $u = \frac{y}{x}$. Then $y = ux \Rightarrow y' = xu' + u$

So, $xu' + u = u + 2x^2 \sin^2(u)$

$$\Rightarrow u' = \frac{2x^2}{x} \sin^2(u)$$

$$\Rightarrow \frac{du}{\sin^2(u)} = 2x dx$$

$$\Rightarrow \int \operatorname{cosec}^2(u) du = \int 2x dx$$

$$\Rightarrow -\cot u = x^2 + C$$

$$\Rightarrow \boxed{-\cot\left(\frac{y}{x}\right) = x^2 + C}$$

$$\left(\text{or } y = x \cot^{-1}(-x^2 - C) \right)$$

Method of Exact ODEs:

Recall: If a function $u(x, y)$ has continuous partial derivatives, then

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

If $u(x, y) = C$ then $du = 0$

Example: $u = x + x^2 y^3 = C$

Then $du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0$

$$\Rightarrow (1 + 2xy^3) dx + (3x^2 y^2) dy = 0$$

$$\Rightarrow y' = - \frac{1 + 2xy^3}{3x^2 y^2}$$

- A first order ODE

$$M(x, y) + N(x, y) y' = 0 \quad \left(\text{or } y' = \frac{dy}{dx} = - \frac{M(x, y)}{N(x, y)} \right)$$

can be rewritten as:

$$M(x, y) dx + N(x, y) dy = 0$$

This ODE is an Exact Differential Equation

if there is a differentiable function

(hence partial derivatives exist and are continuous)

such that

$$\frac{\partial u}{\partial x} = M(x, y), \quad \frac{\partial u}{\partial y} = N(x, y)$$

and hence $M(x, y) dx + N(x, y) dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = du = 0$

↓
(Exact differential)

Ex: $\cos(x+y) dx + [3y^2 + 2y + \cos(x+y)] dy = 0$

Consider $u(x,y) = \sin(x+y) + y^2 + y^3$

Then $\frac{\partial u}{\partial x} = \cos(x+y)$

$$\frac{\partial u}{\partial y} = \cos(x+y) + 2y + 3y^2$$

Then the differential equation becomes

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 \Rightarrow du = 0 \Rightarrow u = C \text{ (constant)}$$

(Implicit solution of
the ODE)

$$\Rightarrow \boxed{\sin(x+y) + y^2 + y^3 = C} \text{ (solution of the ODE)}$$

Solution of Exact ODE :

$$M dx + N dy = du = 0$$

$$\Leftrightarrow u = C \text{ (Implicit solution of ODE)}$$

ie. Level curves of the function u are solutions of this ODE.

Question ①: How to check if an ODE is exact?

Question ②: Once it is exact, how to find its solution?

• An ODE $Mdx + Ndy = 0$ is exact
iff $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Note: Thus if $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, we can conclude
that it is not exact.

Question ② Once we know that $Mdx + Ndy = 0$
is exact, we know that there exists
a function $u(x, y)$ such that

$$M = \frac{\partial u}{\partial x} \quad \text{and} \quad N = \frac{\partial u}{\partial y}$$

To find u , start with $\frac{\partial u}{\partial x} = M(x, y)$

$$\text{Then } u(x, y) = \int \frac{\partial u}{\partial x} dx = \int M(x, y) dx + g(y)$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) + g'(y)$$

Now we know that $\frac{\partial u}{\partial y} = N$

$$\Rightarrow \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) + g'(y) = N(x, y)$$

$$\Rightarrow g'(y) = N(x, y) - \frac{\partial}{\partial y} \left(\int M(x, y) dx \right)$$

Integrate to find $g(y)$

$$\text{Then } \boxed{u(x, y) = \int M(x, y) dx + g(y)}$$

Note: Can also be done by first integrating $N(x, y)$ with respect to y .

Ex: $\cos(x+y) dx + [3y^2 + 2y + \cos(x+y)] dy = 0$

- $M(x, y) = \cos(x+y)$, $N(x, y) = 3y^2 + 2y + \cos(x+y)$

$$\frac{\partial M}{\partial y} = -\sin(x+y), \quad \frac{\partial N}{\partial x} = -\sin(x+y)$$

So, the ODE is exact.

- So, there exists a function $u(x, y)$ such that $\frac{\partial u}{\partial x} = M(x, y) = \cos(x+y)$

$$u(x, y) = \int \cos(x+y) dx$$

$$\Rightarrow u(x, y) = \sin(x+y) + g(y)$$

$$\text{Now } \frac{\partial u}{\partial y} = \cos(x+y) + g'(y)$$

$$\text{Hence } \frac{\partial u}{\partial y} = N(x, y) = \cos(x+y) + g'(y)$$

$$\Rightarrow 3y^2 + 2y + \cancel{\cos(x+y)} = \cancel{\cos(x+y)} + g'(y)$$

$$\Rightarrow g'(y) = 3y^2 + 2y \Rightarrow g(y) = y^3 + y^2 + C$$

$$\text{So, } \boxed{u(x, y) = \sin(x+y) + y^3 + y^2 + C}$$

• can also be done by taking

$$\frac{\partial u}{\partial y} = N(x, y) = 3y^2 + 2y + \cos(x+y)$$

$$\Rightarrow u(x, y) = \int N(x, y) dy = y^3 + y^2 + \sin(x+y) + f(x)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \cos(x+y) + f'(x) \Rightarrow \cos(x+y) = \cos(x+y) + f'(x)$$

$$\Rightarrow f'(x) = 0 \Rightarrow f(x) = C \text{ (some constant)}$$

$$\text{Thus } \boxed{u(x, y) = y^3 + y^2 + \sin(x+y) + C}$$

(Therefore the answers are same.)